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Bias correction of climate models for hydrological modelling – are simple methods still useful?

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ABSTRACT: The spatial climatic characteristics of the Himalayas are complex and a challenge for regional climate models (RCMs). There is no doubt that some form of correction before any application of RCM simulations is a must. In recent years, simple bias correction techniques have been overshadowed by more popular and complex bias correction techniques. In this study an attempt is made to compare the performance of a simple and of a comparatively complex correction technique for hydrological analysis at a monthly resolution in the Kaligandaki River Basin of Nepal. The research workflow consists of bias correction of temperature and precipitation using a simple technique (linear scaling) and a comparatively complex one (quantile mapping). The performance at monthly resolution is evaluated against observed meteorological data while a combined evaluation is made *via* hydrological model response analysis. The wetter and colder RCM estimates were significantly improved after bias correction. The hydrological modelling response also shows the importance of the bias correction of the RCMs. However, no significant difference was observed between the outputs of linear scaling and quantile mapping which exhibited almost identical performances. Hence, this study has a novel conclusion that a simple method, such as linear scaling, is sufficient for hydrological analysis at monthly resolution.

KEY WORDS bias correction; Himalayan basin; hydrological modelling; Kali Gandaki; linear scaling; quantile mapping; SWAT

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1. Introduction

With its extensive cryosphere, the Hindu Kush Karakoram Himalaya region has often been referred to as the third pole (Qiu, 2008) and the water towers of Asia. Ten of the largest Asian rivers originate from the Himalayas alone which provide water to an estimated population of 1.3 billion (Khadka *et al*., 2014). Nestling eight out of the ten tallest mountain peaks of the world, the Nepalese Himalayas rise above the southern plains and hills abruptly, maintaining a staggering topographical gradient. The Nepalese river basins originating from this region have an altitudinal range from near sea level to more than 8000 m above sea level (masl). This results in complex spatial distribution characteristics of precipitation and temperature in the region which adds to the already challenging task of modelling mountainous hydrology.

The impact that climate change poses on a low-latitude mountainous cryosphere like the Himalayas and its resulting climate-driven response on mountainous river hydrology have been well covered by existing literature (Barnett *et al*., 2005; Adam *et al*., 2009; Khadka *et al*., 2014; Panday *et al*., 2014). In an effort to understand the impact of climate change in the future, climate research bodies around the world have been modelling Earth's climate with global climate models (GCMs) which generate meteorological variables by solving the equations of thermodynamics, mass and momentum (Aung *et al*., 2016). Regional climate models (RCMs) may then be used to refine the output spatial resolution which is forced using GCM output as lateral boundary conditions. However, due to systematic and random model errors, GCM simulations often show considerable deviation from observation (Aung *et al*., 2016). By the nature of their dependence on the parent GCM, RCMs inherit a part, if not all, of the bias. For the steep gradient topography of Himalayan basins, estimates of these climate variables can be even more questionable, requiring validation with observed data and use of some form of correction before any application and analysis of climate driven hydrology into the future.

Numerous recent studies have been carried out on the techniques of bias correction and their comparison using different climate estimates of varying resolutions over a range of physiographical regions (Yang *et al*., 2010; Themeßl *et al*., 2011; Berg *et al*., 2012; Gudmundsson *et al*., 2012; Piani and Haerter, 2012; Teutschbein and Seibert, 2012, 2013; Chen *et al*., 2013; Gutjahr and Heinemann, 2013; Teng *et al*., 2015). The techniques vary from simple ones like the delta change technique and linear scaling (LS) to relatively more complex ones such as power transformation, quantile mapping (QM), distribution based scaling and copula based techniques. Most of the studies to date have focused on finding/testing the best technique that brings climatic variables nearest to the ground observations on a daily timescale with comparison including the frequency of wet days, daily intensities and percentile values (Gudmundsson *et al*., 2012; Teutschbein and Seibert, 2012, 2013; Chen *et al*., 2013). Bias correction methods such as QM modify the distribution of rainfall in a way that a simple bias correction method such as LS does not. When the distribution is expected to change (i.e. more extreme rainfall events, changes in wet/dry days) the extra complexity is warranted, and

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so the choice of QM is necessary at finer (e.g. daily or sub-daily) resolutions. Based on this, QM has been heralded as the better bias correction technique (Themeßl *et al*., 2011; Berg *et al*., 2012; Gudmundsson *et al*., 2012; Teutschbein and Seibert, 2012, 2013; Chen *et al*., 2013; Gutjahr and Heinemann, 2013; Teng *et al*., 2015). More recent advancements in bias correction methods comprise copula density and distribution based scaling. Piani and Haerter (2012) dynamically coupled temperature and precipitation to form a 2D probability distribution function or copula density. They concluded that, when temperature and precipitation data are used as input for hydrological models, such inter-variable correlations and higher order statistical links are likely to be important. Yang *et al*. (2010) compared a distribution based scaling method to the simple delta change method wherein it was found that the former identified differences between the two climate projections that were not seen with delta change. However, studies on the usefulness of easy to use, simpler techniques such as LS when the analysis is limited to monthly or seasonal resolution are sparse. In addition, when it comes to analysts involved in water resource studies in the context of climate change, they do not all possess the prerequisite skills for the complex techniques of bias correction such as previous knowledge of higher level statistics and programming skills. In contrast, simpler techniques such as LS are easy to implement, involving only a few steps of calculation in a spreadsheet software. Thus, simple techniques of bias correction are more approachable for a larger number of users in the water resource research community. Their performance in a hydrological setting at coarse temporal resolution needs to be assessed and compared with comparatively complex techniques.

Teng *et al*. (2015) assessed four bias correction techniques applied to the weather research and forecasting model simulated precipitation for eight catchments in southeast Australia and the resulting impact on runoff using two lumped hydrological models, GR4J and Sacramento. Chen *et al*. (2013) used the HSAMI model to analyse seasonal extreme flows as a result of six different bias correction techniques. Similarly, Teutschbein and Seibert (2012) used the HBV hydrological model for analysing the best correction technique in five Swedish catchments which involved analysis of flow extremes. All of this research has concluded that distribution-based techniques are superior to mean-based techniques. However, in Teutschbein and Seibert (2012), out of the five study areas the alpine catchment (Tännfors) showed major deviations in results and gave similar objective function optimization for all techniques. This odd result was attributed to its mountainous characteristics. From their study, Chen *et al*. (2013) also concluded that the performance of bias correction is 'location dependent'. However, hydrological modelling adds a layer of uncertainty to already uncertain climate data. Nevertheless, the uncertainty induced due to hydrological modelling will be cancelled out when comparing the simulated discharge values from two bias correction techniques.

As established in the literature comparing LS and QM, there is little doubt that the latter is necessary for daily scale analysis due to its ability to correct intensity and wet day frequency. However, for water resource applications at monthly resolution, where daily fluctuations average out, LS may also perform well. Thus, the novelty of this study is that it aims to answer the question: can a simple bias correction method yield similar performance to a relatively complex bias correction method for hydrological analyses at monthly resolution? The study takes the challenge of evaluating the usefulness of a simple bias correction technique (LS) against a comparatively complex one (QM) in one of the basins of the Nepalese Himalayas at monthly resolution including hydrological analysis using the Soil and Water Assessment Tool (SWAT) model.

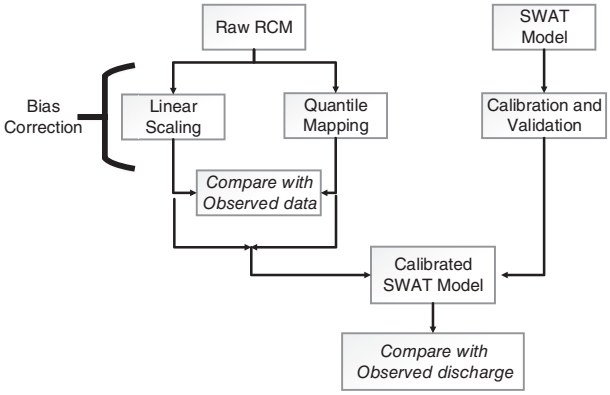


Figure 1. Research methodology.

2. Materials and methods

The main focus of the study is to evaluate the performance of LS and a comparatively complex QM bias correction technique widely applied for RCMs and GCMs and the resulting runoff response using the hydrological model SWAT. Developed at the US Department of Agriculture Agricultural Research Service, the SWAT model is a semi-distributed model that runs on daily time steps to simulate continuous hydrological processes over a long period (Neitsch *et al*., 2011). Both processes, bias correction of climate (precipitation and temperature) data and calibration–validation of the SWAT model, were performed on a daily timescale. However, in order to assess the suitability of LS at monthly resolution, the bias corrected climate data and the resulting simulated discharge data were then aggregated to monthly values for comparison with the observations. The results obtained from the two correction methods and the observations were compared using monthly average plots, hydrographs and probability of exceedence graphs. The overall methodology of study is shown in Figure 1.

2.1. Study area

The Kaligandaki River Basin (KRB) is located in the middle of Nepal (Figure 2). Delineated by the discharge station located at Ansigh-Andhi Ghat, the KRB has a total basin area of 10 666 km^2. The basin accommodates Dhaulagiri (8167 masl) and Annapurna I (8091 masl), the world's 7th and 10th highest mountain peaks respectively. The Kaligandaki River flows between these two mountains along arguably the world's deepest gorge, the Kaligandaki Gorge, which is 5571 m below the Annapurna I peak. In Nepal, a year is subdivided into four seasons. The pre-monsoon (March–May) is followed by a 4 month monsoon (June–September) which in turn is succeeded by a 2 month post-monsoon (October–November). The remaining 3 months from December to February are the coldest and comprise the winter. With a massive elevation range of almost 8000 m (322–8167 masl), the temperature and rainfall vary significantly with space and time in the KRB. The annual rainfall decreases with increase in elevation and varies from as low as 114 mm to as high as 5527 mm. In the KRB, the minimum temperature in the mountains drops to −26 °C in winter and rises to as high as 36 °C in the lower parts of the

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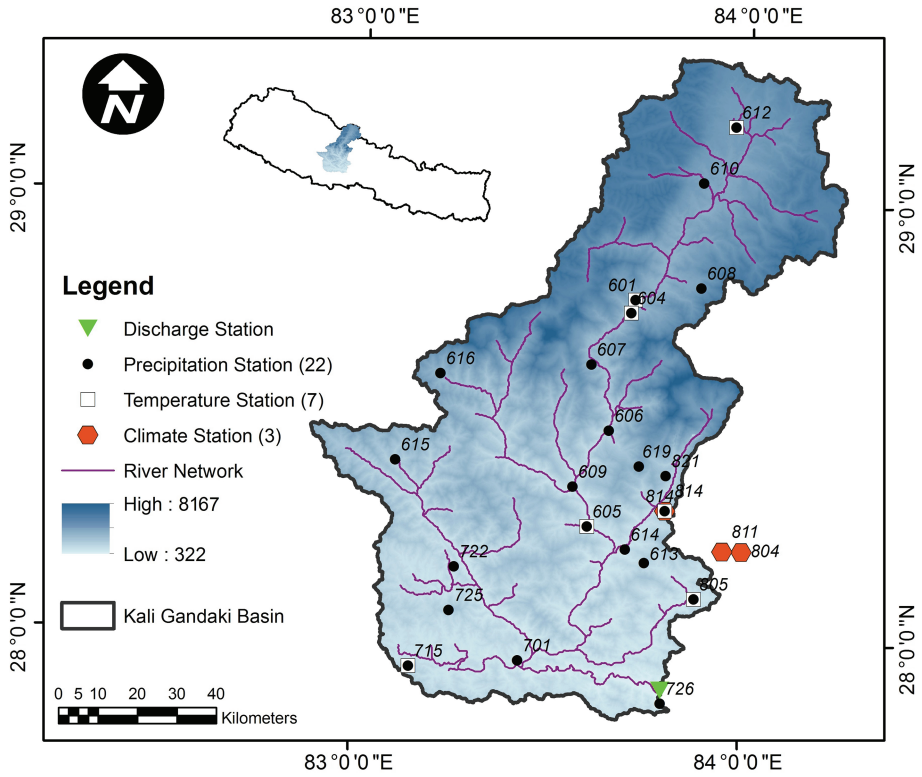


Figure 2. Location map of Kaligandaki River Basin (the numbers in parentheses represent the number of stations). [Colour figure can be viewed at wileyonlinelibrary.com].

basin. The average discharge of Kaligandaki River is 422 $m^3 s^{-1}$ which can reach to more than 5000 $m^3 s^{-1}$ during the monsoon season.

As *per* the land use map developed by the International Centre for Integrated Mountain Development (ICIMOD) for the year 2010 (Uddin *et al*., 2015), 12 types of land use can be found in the study area. The area is dominated by forest, covering nearly 30% of the area. The northern part of the basin is mainly covered by snow and glacier (covering 12.5%) and barren land (covering 23% of the total area). Agriculture, shrub land and grassland dominate the southern part of the basin. Around 24% of the total area is covered by agricultural land and villages and 12% by grass and shrub.

2.2. Data

2.2.1. *Physiographical data*

The basin's physiography data required for the hydrological model included the topographical data, land use data and soil data. The topography was obtained from the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) Digital Elevation Model (DEM) (Tachikawa *et al*., 2011). The land use data were acquired from ICIMOD (Uddin *et al*., 2015) who were the developers of the product, while the soil information was collected from the Soil and Terrain Database (SOTER) Nepal project (Oldeman and Van Engelen, 1993). Brief information and data sources are listed in Table 1.

2.2.2. *Observed climate*

Weather data from 22 precipitation stations, seven temperature stations and three climate stations (measuring relative humidity, solar radiation and wind speed) were collected for 1980–2009 from the Department of Hydrology and Meteorology, Nepal, for the research analysis. All the weather stations except two climate stations lie within the study area. The daily discharge data of the Kaligandaki River at Ansigh-Andhi Ghat were also collected from the Department of Hydrology and Meteorology from 1996 to 2009.

Table 1. Data and corresponding sources.

Data	Source	Spatial/temporal resolution
Topography	ASTER – METI and NASA (http://reverb.echo.nasa.gov/reverb)	30 m × 30 m
Land use	International Centre for Integrated Mountain Development (ICIMOD)	30 m × 30 m/year 2010
Soil map	Soil and Terrain Database (SOTER)	1:1 000 000
Observed hydrometeorology	Department of Hydrology and Meteorology, Nepal	Point dataset/daily

2.2.3. *RCM data*

Daily precipitation, maximum and minimum temperature from three RCMs were analysed for 1980–2005. These RCMs were downscaled for the South Asian region and were therefore selected for this study. The selected RCMs have been proved to be suitable for Himalayan basins in Nepal (Aryal, 2016). The RCM data were downloaded from the South Asia CORDEX data

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portal (http://cccr.tropmet.res.in/home/index.jsp) and have a resolution of 0.5°. The name and parent GCM (in parentheses) of the RCMs used in the study are CCAM (ACCESS1.0), CCAM (CNRM-CM5) and CCAM (MPI-ESM-LR).

2.3. Bias correction techniques

Even though the RCMs and GCMs show good results in mimicking the observed data on a larger scale, they still have some bias on studying at basin level (Shrestha *et al*., 2016). The bias correction technique aims to correct the mean, variance and/or quintile of the model time series variable using a certain correction factor so that the corrected model time series matches closely with the observed variable. In this study two bias correction techniques, LS and QM, are used to correct the biases of RCMs on a daily timescale.

2.3.1. Linear scaling (LS)

LS is the simplest bias correction technique applied in various studies (Ines and Hansen, 2006; Teutschbein and Seibert, 2012; Shrestha *et al*., 2015) and is used to adjust the RCM mean value. The difference between the monthly mean observed and model data is applied to the model data to obtain bias corrected climate data. Data analysis relying on LS bias correction (V.1.0) (Shrestha, 2015) was used for the application of this method. The following equations were used for the LS technique:

$$P_{\text{his}}(d)^{*} = P_{\text{his}}(d) \cdot \left[\mu_{\text{m}}\left\{P_{\text{obs}}(d)\right\} / \mu_{\text{m}}\left\{P_{\text{his}}(d)\right\}\right] \quad (1)$$

$$P_{\text{sim}}(d)^{*} = P_{\text{sim}}(d) \cdot \left[\mu_{\text{m}}\left\{P_{\text{obs}}(d)\right\} / \mu_{\text{m}}\left\{P_{\text{his}}(d)\right\}\right] \quad (2)$$

$$T_{\text{his}}(d)^{*} = T_{\text{his}}(d) + \left[\mu_{\text{m}}\left\{T_{\text{obs}}(d)\right\} - \mu_{\text{m}}\left\{T_{\text{his}}(d)\right\}\right] \quad (3)$$

$$T_{\text{sim}}(d)^{*} = T_{\text{sim}}(d) + \left[\mu_{\text{m}}\left\{T_{\text{obs}}(d)\right\} - \mu_{\text{m}}\left\{T_{\text{his}}(d)\right\}\right] \quad (4)$$

where P is precipitation, T is temperature, d stands for daily, μ_{m} is the long term monthly mean, an asterisk indicates bias corrected, his refers to historical raw RCM data, obs stands for observed data and sim is the raw RCM future data.

2.3.2. Quantile mapping (QM)

The main objective of QM is to correct the quantiles of RCM data to match the quantiles of observed data by creating a transfer function to shift the quantiles of precipitation and temperature. Distribution based QM (Gudmundsson *et al*., 2012; Teutschbein and Seibert, 2012) as well as empirical QM (Gudmundsson *et al*., 2012) are used in correcting precipitation and temperature. In this study, empirical QM was used with the 99-percentile table generated and linear interpolation between them. The QM method was implemented in R language (Venables and Smith, 2012) using package 'qmap' (Gudmundsson, 2014). The following equations were used for the QM technique:

$$P_{\text{his}}(d)^{*} = F^{-1}_{\text{obs,m}}\left\{F_{\text{his,m}}\left(P_{\text{his,m}}\right)\right\} \quad (5)$$

$$P_{\text{sim}}(d)^{*} = F^{-1}_{\text{obs,m}}\left\{F_{\text{sim,m}}\left(P_{\text{sim,m}}\right)\right\} \quad (6)$$

$$T_{\text{his}}(d)^{*} = F^{-1}_{\text{obs,m}}\{F_{\text{his,m}}(T_{\text{his,m}}\}] \quad (7)$$

$$T_{\text{sim}}(d)^{*} = F^{-1}_{\text{obs,m}}\left\{F_{\text{sim,m}}\left(T_{\text{sim,m}}\right)\right\} \quad (8)$$

where F is the cumulative distribution function and F^{-1} is its inverse.

2.4. SWAT modelling

A physically based semi-distributed hydrological model SWAT was used to model runoff from the Kaligandaki River. The inputs needed for the modelling are spatial data, i.e. the DEM, land use and soil maps along with physical properties and weather data, i.e. precipitation, temperature, relative humidity, solar radiation and wind speed. Shrestha *et al*. (2016) found SWAT to perform very successfully (co-efficient of determination R^2 of 0.83, 0.90, and Nash–Sutcliffe efficiency (NSE) of 0.81, 0.87 (Table 2)) in the Indrawati basin (east of Kaligandaki) of Nepal. Another study at a larger basin scale in the Koshi River (easternmost Nepalese Himalayan basin) portrayed SWAT to be an outstanding hydrological model with values for R^2 of 0.98, 0.88 and NSE of 0.97, 0.87 (Devkota and Gyawali, 2015). So, together with this study previous hydrological studies have also confirmed SWAT to be suitable for Nepalese basins.

The model was set up with 103 sub-basins and 479 hydrological response units. Each sub-basin is divided into five elevation bands. The sub-basins then extract the data from the nearest station and calculate the values for each of the elevation bands using lapse rates for temperature (TLAPS, °C km^{-1} rise in elevation) and precipitation (PLAPS, mm km^{-1} rise in elevation). This helped in achieving heterogeneity in precipitation and temperature which is representative and characteristic of the KRB. For routing the generated discharge in the model, the SCS curve number procedure of SWAT was used. The model was then calibrated against the observed discharge at the Ansigh-Andhi Ghat station on the Kaligandaki River. Daily discharge data from 1996 to 2005 and from 2006 to 2009 were used as calibration and validation periods respectively. The model performance was evaluated using both graphical and statistical parameters. Coefficients of determination (R^2), NSE, relative Nash–Sutcliffe efficiency (NSE_{rel}), percentage bias (PBIAS) and ratio of the root mean square error to standard deviation (RSR) were used to check the accuracy of the model. Descriptions and formulae of each of the statistical parameters are shown in Table 2. The threshold of goodness of fit for the model was based on Moriasi *et al*. (2007). The calibration and validation of SWAT was efficiently achieved with the above mentioned statistics employing SWAT Calibration Helper v1.0 (Shrestha, 2016) which is a customized Visual BASIC for Applications coded Microsoft Excel file.

3. Results

3.1. Performance evaluation of bias correction on RCM simulations

Mean monthly values of RCM simulations before and after correction against observed values are illustrated in Figure 3 (maximum and minimum temperature) and Figure 4 (precipitation). The time range used for averaging out monthly values is 1996–2005. However, it should be noted that the correction functions for both bias correction techniques were generated using the historical time range from 1980 to 2005.

All the RCMs significantly underestimated the temperature, i.e. gave much colder values for both daily extremes. At

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Table 2. Descriptions and formulae of statistical parameters.

Formula	Description
$R^2 = \left(\frac{n\sum Q_i Q'_i - \sum Q_i \sum Q'_i}{\left(\sqrt{n(\sum Q_i^2) - (\sum Q_i)^2}\right) \times \left(\sqrt{n(\sum Q_i'^2) - (\sum Q'_i)^2}\right)} \right)^2$	R^2 provides the strength of relation between observed and simulated values. Its value ranges from 0 to 1, a value close to 0 means very low correlation whereas a value close to 1 represents high correlation between observed and simulated discharge.
$NSE = 1 - \frac{\sum_{i=1}^{n} (Q_i - Q'_i)^2}{\sum_{i=1}^{n} \left(Q_i - \overline{Q_i}\right)^2}$	NSE determines the relative magnitude of the residual variance compare to the measured data variance (Moriasi *et al*., 2007). Its value ranges from -∞ to 1, where 1 indicates a perfect model and a value less than 0 indicates that the mean value of the observed time series would have been a better predictor than the model.
$NSE_{rel} = 1 - \frac{\sum_{i=1}^{n} \left(\frac{Q_i - Q'_i}{Q_i}\right)^2}{\sum_{i=1}^{n} \left(\frac{Q_i - \overline{Q_i}}{\overline{Q_i}}\right)^2}$	It has greater reaction to low flows and is insensitive to peaks. Ideal statistic to evaluate base flow simulation performance of hydrological models. Its value ranges from -∞ to 1. Values near to 1 indicate good simulation of low flows (baseflow).
$PBIAS = \frac{\sum_{i=1}^{n} (Q_i - Q'_i) * 100}{\sum_{i=1}^{n} (Q_i)}$	It measures the difference between the simulated and observed quantity and its optimum value is 0. A positive value of the model represents underestimation whereas a negative value represents the model overestimation.
$RSR = \frac{RMSE}{STDEV_{obs}} = \frac{\sqrt{\frac{\sum_{i=1}^{n} (Q_i - Q'_i)^2}{n}}}{\sqrt{\frac{\sum_{i=1}^{n} \left(Q_i - \overline{Q_i}\right)^2}{n}}}$	The lower value of RMSE (Root Mean Square Error) is commonly acceptable and one of the widely used error parameters. However, the satisfactory threshold of RMSE is case specific. Therefore, RSR is chosen as a complementary indicator to RMSE. The optimum value of RSR is 0 and a higher value indicates lower model performance.

Jomsom (2744 masl) both minimum and maximum temperatures were underestimated by approximately 20 °C during winter and by around 10–15 °C during the middle of the monsoon. A similar pattern of estimation was observed at the low altitude stations of Baglung and Syangja also, except at the pre-monsoon in Syangja where the maximum temperature was slightly overestimated. However, all three RCMs showed a small spread.

For temperature, the bias in RCM temperature is notably reduced to match the observed values. All three bias corrected RCM temperatures are close to each other with a smaller spread between them. Also, both the correction techniques show a similar performance with almost identical spread.

The RCMs show significant variability and bias throughout the year compared to the observed mean monthly rainfall. However, the variability seems to be comparatively lower during the pre-monsoon period and higher during the monsoon. It can also be noted that the RCMs generally overestimate precipitation, even more so at lower elevations (stations 605, 726) compared to higher elevations (stations 615, 616). Furthermore, the onset of the monsoon is earlier in RCM simulations by a month compared to the observed data. Such an effect is not evident for the end of the monsoon.

On application of the two bias correction techniques, the wetter RCM estimates were improved bringing the mean monthly values closer to the observed values. Moreover, both techniques show similar performance with the shaded areas almost overlapping each other at each of the four stations. The early onset of the monsoon as observed in the RCM precipitation simulations was also adjusted to match the observed data.

3.2. Hydrological modelling using observed climate

The model was able to capture the observed discharge of the KRB outlet at Ansigh-Andhi Ghat as shown by its simulation performance. The basin outlet discharge simulation performance statistics are listed in Table 3. At daily resolution, both calibration and validation periods show very good values of R^2, NSE, NSE_{rel} and RSR and all the values lie within the very good rating based on Moriasi *et al*. (2007) with R^2 and NSE greater than 0.7 and RSR less than 0.6. Similarly, for both calibration and validation periods the PBIAS lies within the range ±25%. Hence, with observed climatic forcing, the SWAT model was able to simulate the mountainous river hydrology of the KRB.

3.3. Hydrological modelling using bias corrected RCM simulations

After evaluation of bias correction techniques on RCM forcings (precipitation and temperature separately), the combined effect of the bias correction of these climatic variables on simulated discharge was analysed. Hydrological simulation with the raw RCM shows very high overestimation of discharge throughout the year (see the monthly average graph of Figure 5). In the figure, 'Simulated' represents the model discharge with observed rainfall and temperature, 'Raw' represents the model discharge with raw RCM data, 'LS' represents the model discharge with LS bias corrected RCM data and 'QM' represents the model discharge with QM bias corrected RCM data. This can be accredited to the substantial positive bias in RCM precipitation and the reduced evapotranspiration due to high

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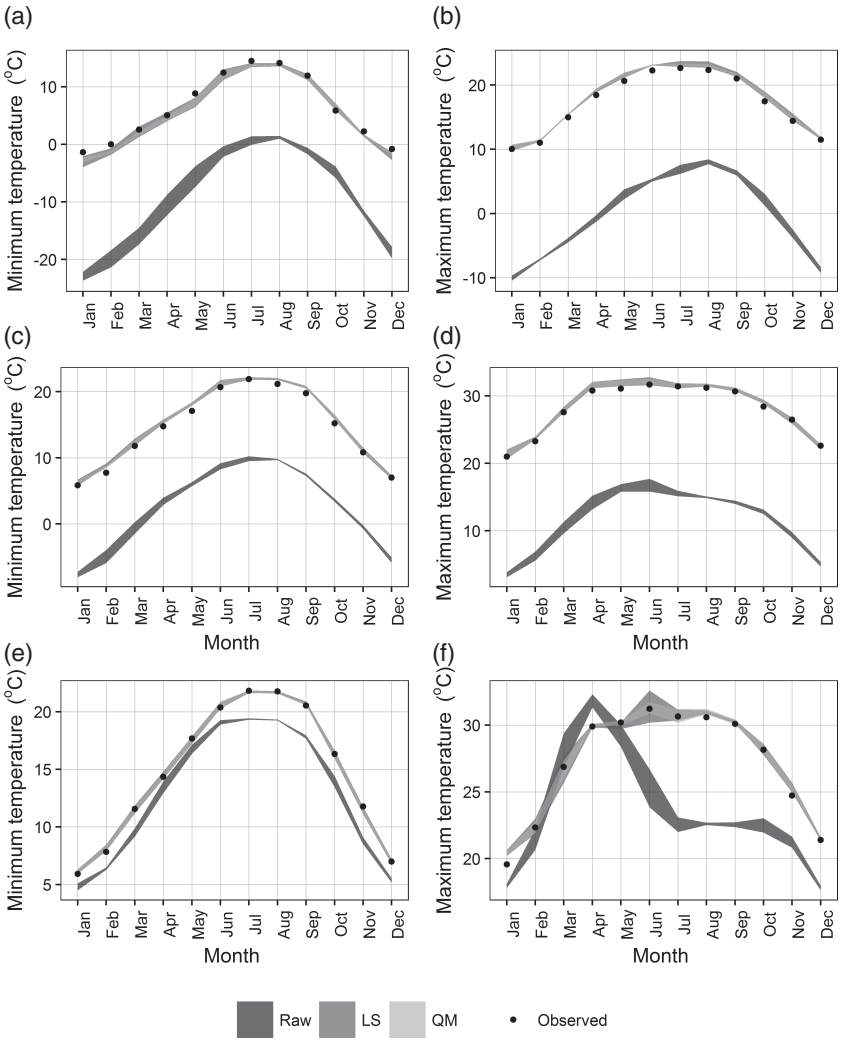


Figure 3. Variability in monthly temperatures across regional climate models (RCMs) before and after bias correction: high altitude station at Jomsom (601) (a, b), low altitude stations at Baglung (605) (c, d) and Syangja (805) (e, f), minimum temperature (a, c, e), maximum temperature (b, d, f). Variability due to RCMs: raw, corrected by linear scaling (LS) and corrected by quantile mapping (QM).

negative bias in RCM temperatures. However, with bias corrected RCM input, the model simulations improved greatly and fitted the observed discharge well. The monthly simulations for winter and pre-monsoon show close agreement with the observed values and a good agreement was achieved during post-monsoon as well. No significant difference was detected between the model discharges under the two correction techniques. The probability of exceedence of the monthly discharge (Figure 5(b)) was prepared using the monthly averaged values of discharge. However, in order to preserve visibility, the graph shows points at each 5% increment in exceedence. The graph reveals that the correction techniques improve the distribution of discharge lower than Q25 (flow exceeded only 25% of the time). For higher discharge, despite the deviation from observed discharge values, the largely overestimating RCM forced discharge was significantly revised and thus comparable to model output with the observed climate.

4. Discussion

The hydrological performance of the SWAT model for the KRB was notable. The 0.70+ statistics for all R^2, NSE and NSE_{rel} show the model to have fitted the pattern, peaks and base flow of the hydrograph well.

In general, the current study shows significant overestimation in precipitation and underestimation in temperature estimates of the three RCMs used, i.e. wetter and colder climate estimates. This was also reflected in the hydrological response of the SWAT model (Table 4) wherein large discharge overestimation occurred due to high precipitation and low evapotranspiration. Studies such as this one, with RCM estimates based on the latest representative concentration pathway scenarios under the International Panel for Climate Change Fifth Assessment Report (IPCC AR5), are rare in the Nepalese Himalayas. Of the few, in Shrestha *et al*. (2015), Hadley Centre Global Environmental Model version 3 regional climate model (HadGEM3-RA RCM) also exhibits overestimation of mean precipitation (by almost a factor of 2) at four out of six precipitation stations. An underestimation of maximum and minimum temperature by around 9 and 2 °C respectively was also observed in the same study. In addition, in RCM precipitation estimates, the present study demonstrates high variability amongst the RCMs as well, especially during the monsoon. The available studies and results thus suggest adjustment in RCM simulations before any application.

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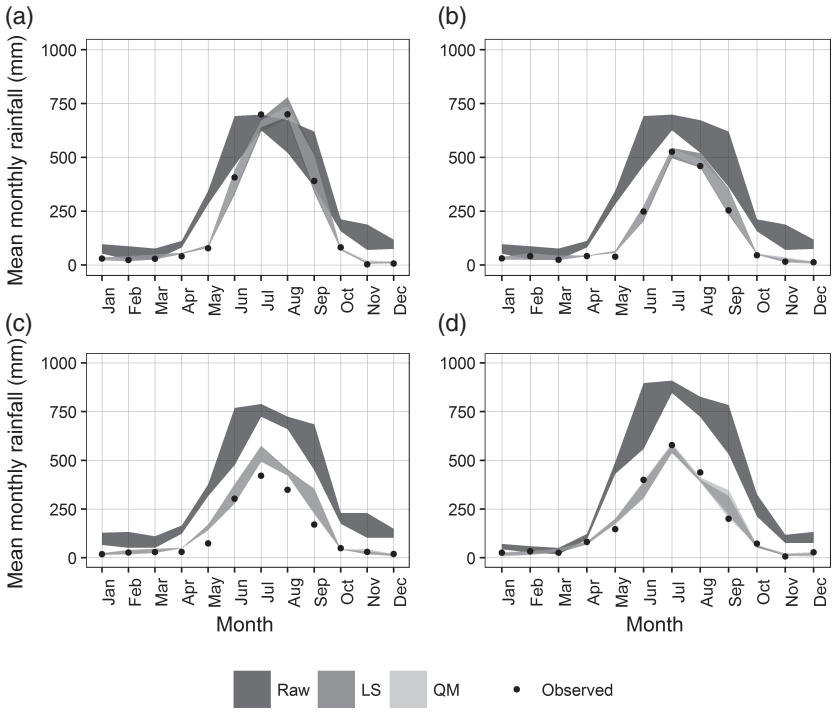


Figure 4. Variability in monthly precipitation across regional climate models (RCMs) before and after bias correction. High altitude stations at Bobang (615) and Gurja Khani (616) (a, b) and low altitude stations at Baglung (605) and Garakot (726) (c, d). Variability due to RCMs: raw, corrected by linear scaling (LS) and corrected by quantile mapping (QM).

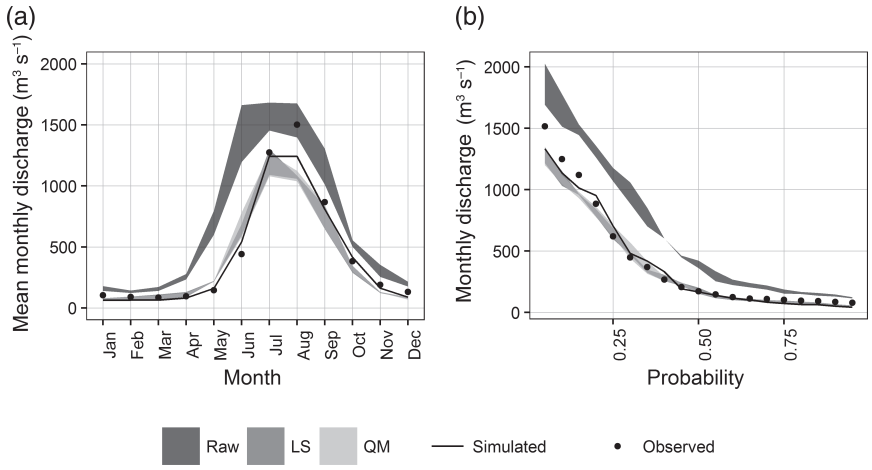


Figure 5. Variability in monthly discharge at basin outlet caused by regional climate model (RCM) forcings before and after bias correction as monthly average (a) and probability of exceedence (b). Variability due to RCMs: raw, corrected by linear scaling (LS) and corrected by quantile mapping (QM).

Table 3. Goodness of fit of the Soil and Water Assessment Tool model for Ansigh-Andhi Ghat station at daily resolution.

Model period	R^2	NSE	NSE_{rel}	PBIAS (%)	RSR
Calibration (1996–2005)	0.84	0.83	0.91	−6.9	0.41
Validation (2006–2009)	0.78	0.70	0.73	+19.7	0.55

Table 4. Statistics (monthly) for model hydrology for various precipitation and temperature model input.

Statistic	Performance for different weather input			
	Simulated	Raw RCM	RCM-LS	RCM-QM
R^2	0.92	0.68	0.78	0.84
NSE	0.91	0.35	0.77	0.82
NSE_{rel}	0.93	−1.30	0.81	0.85
PBIAS	−6.90	+54.10	−10.80	−10.00
RSR	0.31	0.80	0.48	0.42

RCM-LS, linear scaling corrected RCM; RCM-QM, quantile mapping corrected RCM.

Bias correction techniques helped to improve both the magnitude and variability of the RCMs. In the case of precipitation, the 1 month error at the onset of the monsoon was also revised. Previous bias correction studies have focused mainly on finding the best technique with analyses based on

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metrics on hydrometeorological extremes and daily characteristics (Gudmundsson *et al*., 2012; Teutschbein and Seibert, 2012, 2013; Chen *et al*., 2013; Teutschbein and Seibert, 2013). Distribution based techniques (QM) have been heralded as a superior bias correction technique (Themeßl *et al*., 2011; Berg *et al*., 2012; Gudmundsson *et al*., 2012; Teutschbein and Seibert, 2012, 2013; Chen *et al*., 2013; Gutjahr and Heinemann, 2013; Teng *et al*., 2015). In the monthly analysis in this study it was observed that, irrespective of elevation, both techniques gave the same performance. The overall impression from the hydrological output of the two techniques (Table 4) is that there exists little or no difference between the performances of the two techniques. Since most of the analysis in climate change impact is performed on a monthly or seasonal scale rather than daily (except for flood analysis), it was found from this study that mean based techniques like LS (which are simple to learn and use) are equally effective when it comes to analysis at coarser temporal resolution. However, if climate data at daily resolutions are to be analysed, relatively complex methods such as QM are inevitable as the method is able to correct the intensity and wet and dry day frequency of the climate model in addition to the mean.

5. Conclusions

This study confirms the Soil and Water Assessment Tool model to be suitable for hydrological analysis in the Nepalese Himalayan basins at monthly (and daily) resolution. The present study shows RCM simulations to have large discrepancies from observations in the geographical region and marks the necessity for previous correction before application.

With most recent studies pointing out the superiority of complex bias correction techniques, easy to learn techniques such as linear scaling are in the shadow of popular contemporary techniques. However, this study has a novel conclusion that linear scaling is as effective as quantile mapping when it comes to analysis at coarser temporal resolution irrespective of elevation.

It is known that climate change and climate change analysis is a cross-cutting research field wherein researchers from diverse backgrounds need to analyse climate model data and in the process remove their bias. In this regard, the prerequisites of quantile mapping (including previous advanced knowledge of statistics and the programming language) make it a much more complex 'learn and apply' method compared to linear scaling, which can be performed using a simple spreadsheet with a basic knowledge of averaging out the errors. Therefore, simple techniques offer dual benefit, i.e. easy to learn, enveloping a larger research community, and saving resources, including time and effort. Thus, at coarse resolution it can be argued that simple bias correction techniques can be more beneficial.

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